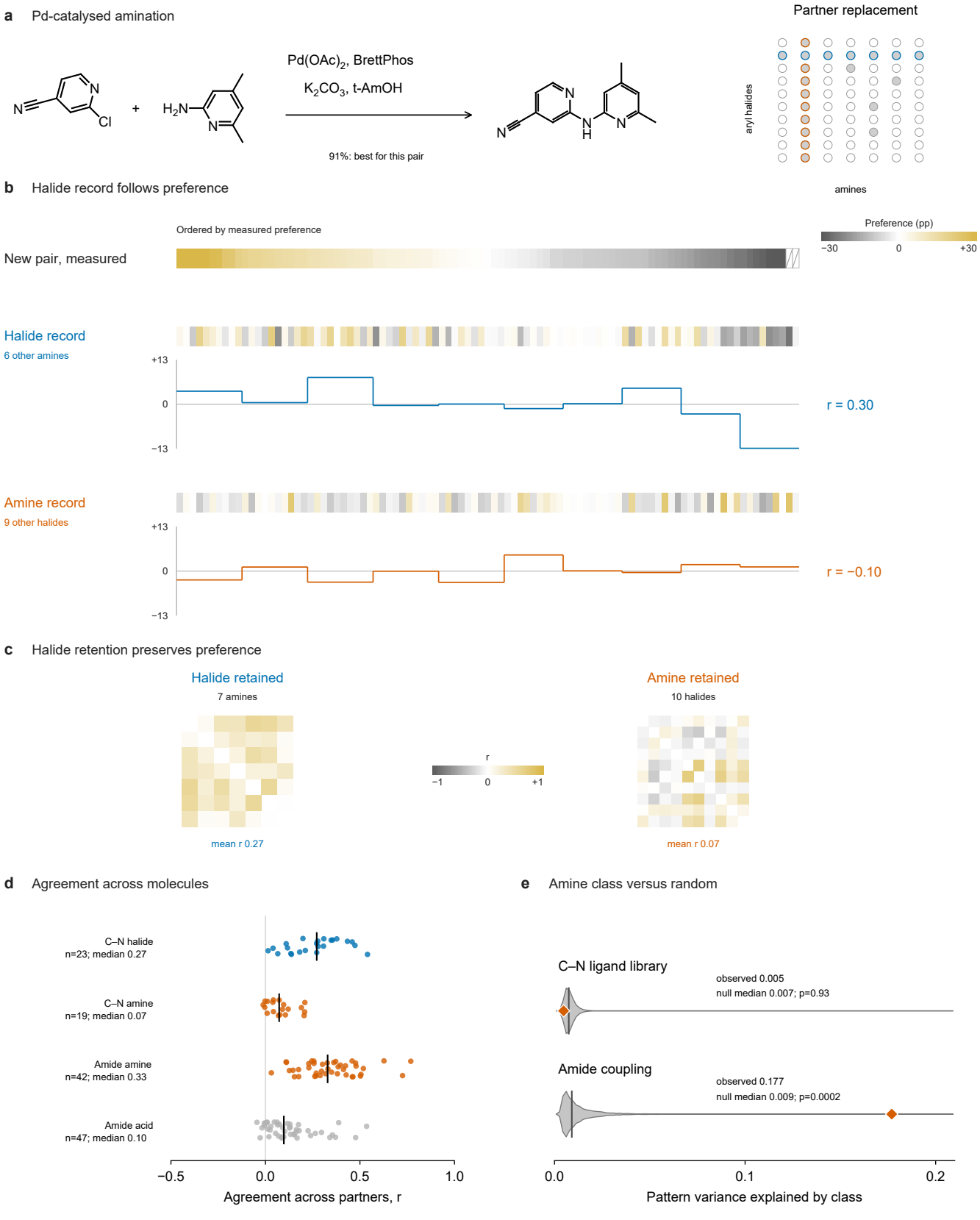

a
Pd-catalysed amination
Pd(OAc)2, BrettPhos
K2CO3, t-AmOH
91%: best for this pair
Partner replacement
aryl halides
amines
b
Halide record follows preference
Ordered by measured preference
Preference (pp)
−30
0
+30
New pair, measured
Halide record
6 other amines
+13
0
−13
r = 0.30
Amine record
9 other halides
+13
0
−13
r = −0.10
c
Halide retained
7 amines
mean r 0.27
r
−1
0
+1
Amine retained
10 halides
mean r 0.07
d
Agreement across molecules
C–N halide
n=23; median 0.27
C–N amine
n=19; median 0.07
Amide amine
n=42; median 0.33
Amide acid
n=47; median 0.10
−0.5
0.0
0.5
1.0
Agreement across partners, r
e
Amine class versus random
C–N ligand library
observed 0.005
null median 0.007; p=0.93
Amide coupling
observed 0.177
null median 0.009; p=0.0002
0.0
0.1
0.2
Pattern variance explained by class